# ADITYA RAJ

Bengaluru, India · adiwildinn@gmail.com · github.com/adityarghh
linkedin.com/in/aditya-raj-79a53b314 · portfolio-aditya-seven-rho.vercel.app

## SUMMARY

AI-focused software developer building and shipping production systems beyond coursework — including a live, deployed product.
Experienced across the full stack: frontend, serverless backend, LLM API integration, distributed systems, and applied ML pipelines.

## EDUCATION

**Vellore Institute of Technology** *2025 – 2029*
*B.Tech in Artificial Intelligence*
Coursework: Data Structures · OOP · Machine Learning · Distributed Systems · Networking · Cyber Risk & Compliance

## TECHNICAL SKILLS

**Languages:** Java, Python, C++, JavaScript
**Frontend:** React, Vanilla JavaScript, TailwindCSS, Responsive UI
**Backend & Systems:** Flask, REST APIs, Serverless (Vercel), TCP/IP Networking, Concurrency
**AI / ML:** Scikit-learn, NumPy, Pandas, NLP, Semantic Search, LLM API Integration
**Security:** API key isolation, defensive JSON parsing, secure serverless deployment
**Tools:** Git, GitHub, VS Code, Vercel

## FLAGSHIP PRODUCT — LIVE DEPLOYMENT

**Baby Steps to the Mic — AI Public Speaking Coach** *JavaScript · Web Speech API · Claude API · Vercel*
*Independently designed, built, and deployed a production AI coaching platform — currently live.*

- Built a Vanilla JS frontend with Web Speech API integration enabling real-time speech transcription and dynamic speaking topic generation per session
- Integrated Claude API via a Vercel serverless backend to deliver structured per-session feedback: scores, strengths, weaknesses, and coaching recommendations
- Implemented defensive JSON parsing to safely handle malformed or unpredictable LLM responses without crashing the session pipeline
- Isolated API keys server-side within serverless functions to prevent client-side exposure; managed environment configuration and production deployment end-to-end
- Made and executed all product decisions — architecture, frontend UX, backend API design, security posture, and deployment — from first commit to production

## PROJECTS

**Distributed Chat System** *Java · TCP/IP · Concurrency*

- Built a multi-client TCP socket server in Java; assigned a dedicated thread per connection and synchronized shared state across concurrent sessions
- Implemented TCP/IP socket programming from scratch: server binds, accepts connections, and routes messages across active client sessions
- Applied thread synchronization primitives to prevent race conditions across shared data structures under simultaneous writes

**DocuQuery — AI Document Retrieval System** *Python · NLP · IR*

- Built an NLP-based document querying pipeline with structured extraction and relevance-ranked retrieval across multi-format inputs
- Built a preprocessing and indexing layer to normalize, tokenize, and vectorize documents into a queryable representation
- Tuned filtering and ranking logic to improve precision of returned results given noisy or ambiguous queries

**Offroad-Semantic — ML Semantic Analysis** *Python · Scikit-learn · NLP*

- Built a semantic classification pipeline: text ingestion, cleaning, TF-IDF feature extraction, vectorization, and Scikit-learn model inference
- Designed semantic mapping workflows to interpret context across unstructured inputs and assign structured categorical labels
- Tested multiple Scikit-learn classifiers and preprocessing combinations to identify the highest-accuracy configuration for the dataset

**PassSecure — Flask Password Security Tool** *Python · Flask · JavaScript*

- Built a Flask web app that evaluates password strength using entropy estimation, pattern matching (dictionary words, sequences, repeated chars), and character class checks
- Implemented secure input handling on both frontend and backend; designed responsive UI with real-time strength indicator and actionable feedback

**Bank Management Simulation** *Java · OOP*

- Designed a modular banking system with Account, Transaction, and Customer classes applying inheritance, encapsulation, and polymorphic dispatch
- Implemented deposit, withdrawal, transfer, and balance inquiry flows with structured transaction logging and edge-case validation

**Library Management System** *Java · SQL · CRUD*

- Built a Java application with full CRUD operations against a relational database: book inventory, member registration, issue/return tracking, and fine calculation
- Separated SQL data access from business logic into distinct layers, keeping the codebase modular and schema changes isolated

## EXPERIENCE

**Core Member — Bit By Bit Club, VIT** *Dec 2025 – Present*

- Drive technical initiatives across collaborative builds; conduct peer code reviews focused on architecture quality and implementation decisions
- Participate in AI/ML-focused discussions and workshops bridging coursework concepts to applied engineering practice

**Design & Content Creator — PranSpanda Club, VIT** *Nov 2025 – Dec 2025*

- Led visual communication campaigns applying UI/UX principles to improve content structure, readability, and audience engagement across club platforms

**Event Anchor & Coordinator** *2023 – 2024*

- Anchored 5+ large-scale academic and cultural events; directed volunteer teams of 15+ and managed live audiences exceeding 500 attendees
- Developed structured public presentation and cross-functional coordination skills under real-time constraints

## CERTIFICATIONS & ACHIEVEMENTS

- **Hackathon — Dawn of Code, VIT (MERN Stack Club):** shipped a full-stack prototype under time constraints
- **AWS Industry Workshop — Amazon Q in Code & Play:** hands-on cloud-native AI developer tooling
- **Machine Learning Foundations Certification** · **Cyber Risk & Compliance Masterclass**
- National-Level Financial Literacy Quiz Participant · International Academic Engagement — VLU Global Talk